

Regularity criterion for a 3D Tropical Climate Model with Damping

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Joint work with

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3D Tropical Climate Model with Damping

We consider the following Cauchy problem:

$$\begin{cases} \partial_t u + (u \cdot \nabla) u - \Delta u + |u|^{\alpha-1} u + \nabla \pi + \operatorname{div} (v \otimes v) = 0, \\ \partial_t v + (u \cdot \nabla) v - \Delta v + |v|^{\beta-1} v + (v \cdot \nabla) u + \nabla \theta = 0, \\ \partial_t \theta + (u \cdot \nabla) \theta - \Delta \theta + \operatorname{div} v = 0, \\ \operatorname{div} u = 0, \\ (u, v, \theta)|_{t=0} = (u_0, v_0, \theta_0), \end{cases} \quad (1)$$

where, for $x \in \mathbb{R}^3$ and $t \geq 0$,

$u = (u_1(x, t), u_2(x, t), u_3(x, t))$ barotropic mode of the velocity field,

$v = (v_1(x, t), v_2(x, t), v_3(x, t))$ first baroclinic mode of the velocity field,

$\pi = \pi(x, t)$ pressure,

$\theta = \theta(x, t)$ temperature (typical midtropospheric temperature).

Here, $v \otimes v$ is the 2-tensor with entries $\{v_i v_j\}_{i,j=1,2,3}$, $\partial_t + (u \cdot \nabla)$ is the *convective derivative along u* and $\alpha, \beta \geq 1$ are the *damping coefficients*.

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Some remarks.

- ▶ Without the damping terms, we recover the usual TCM. When, moreover,
 - ▶ $\theta = 0$, and $\operatorname{div} v = 0$, then (1) is called *viscous Magneto Hydro Dynamics* (MHD) equations.
 - ▶ When also $v = 0$, we recover the classical viscous *Navier-Stokes* (NS) equations for incompressible flow.
- ▶ For simplicity, we have set equal to 1 the multiplying constants in front of $-\Delta u$, $-\Delta v$, $-\Delta \theta$, $|u|^{\alpha-1} u$ and $|v|^{\beta-1} v$. Nothing changes if one scales by positive constants.

A few words on the model

- ▶ In the context of **meteorology**, in order to provide simplified equations to **model the interaction of large scale flow fields and tropical precipitation**, Frierson, Majda, Pauluis (2004) proposed (in 2D) the standard TCM, that is (1) without the diffusion and the damping terms.
- ▶ Roughly speaking, approximating the primitive equations, the model there consists of a **shallow water equation for the velocity field U** in the tropics, and an **equation for moisture (humidity)**.
- ▶ From a vertical decomposition of U , they result defined barotropic mode u and baroclinic mode of U . The first baroclinic mode v is nothing else but the truncation up to the first term of a Galerkin expansion of the baroclinic mode.
- ▶ Usually the barotropic mode appears when $\rho = \rho(\pi)$, while the first baroclinic mode had been used largely in literature about tropical atmosphere dynamics (see e.g. Matsuno (1966) and Gill (1980)).

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(A very short list of) main previous results about regularity

- In **2D** no damping is needed:
 - ▶ **Li and Titi (2016)** proved the **global well-posedness** of (strong) solutions of the Cauchy problem, when the system considered has viscosities in the momentum equations, but no diffusivity in the temperature equation.
 - ▶ **Wan (2016)**: **global well-posedness under small initial data**, for (TCM) with diffusion only on the first baroclinic mode of the velocity.
- For the **3D** case, the question of global regularity for smooth solutions remains generally open.
 - ▶ **Ma, Jiang, and Wan (2016)** considered TCM with **fractional velocity diffusion** and provided local well-posedness of solutions.
 - ▶ In case of (large enough) **fractional diffusion on barotropic mode** **Zhu (2018)** gave the global well-posedness of strong solutions, provided the initial data $(u_0, v_0, \theta_0) \in W^{3,2}(\mathbb{R}^3)$.

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Previous results - 3D TCM with damping

So, it emerges the necessity of damping terms:

- ▶ **Yuan and Chen (2021)** introduced (1) with a damping only on the barotropic mode of the velocity. The **global existence and uniqueness** of a strong solution is established provided $\alpha \geq 4$.
- ▶ **Yuan and Zhang (2021)** considered (1) with damping also on θ .
 - ▶ Global regularity assuming that one of the following three conditions holds true:

$$\alpha, \beta \geq 4,$$

$$\frac{7}{2} \leq \alpha < 4, \quad \beta \geq \frac{5\alpha + 7}{2\alpha}, \quad \gamma \geq \frac{7}{2\alpha - 5},$$

$$3 < \alpha \leq \frac{7}{2} \quad \beta, \gamma \geq \frac{7}{2\alpha - 5},$$

where $\gamma > 1$ refers to an extra damping term in $(1)_3$, i.e. $|\theta|^{\gamma-1} \theta$.

Our contribution

In order to study sufficient conditions for the regularity of local solutions of the Cauchy problem associated with (1), in collaboration with Bisconti and Catania, we established a regularity criterion in terms of homogeneous Besov spaces, in the case $3 < \alpha, \beta < 4$.

The results are contained in a work submitted very recently:

- ▶ [B., Bisconti, Catania, submitted May 2022, preprint online on *arXiv*].

Notations on spaces

Lebesgue spaces For $p \geq 1$, we denote with $L^p = L^p(\mathbb{R}^3)$, endowed with usual norm $\|\cdot\|_p$. In the case $p = 2$, we use the notation $\|\cdot\| = \|\cdot\|_2$.

Sobolev spaces For $s > 0$, we denote the Sobolev spaces $H^s = W^{s,2}(\mathbb{R}^3)$, equipped with norm $\|\cdot\|_{H^s}$. Here $\dot{H}^s$ denotes the standard homogeneous Sobolev space with norm $\|\cdot\|_{\dot{H}^s}$. In terms of the Fourier transform:

$$\text{homogeneous Sobolev space } \dot{H}^s \quad \|f\|_{\dot{H}^s}^2 = \int_{\mathbb{R}^3} |\xi|^{2s} |\hat{f}(\xi)|^2 d\xi < +\infty,$$

and

$$\text{Sobolev space } H^s \quad \|f\|_{H^s}^2 = \int_{\mathbb{R}^3} (1 + |\xi|)^{2s} |\hat{f}(\xi)|^2 d\xi < +\infty,$$

and, when $s > 0$, H^s is equal to $\dot{H}^s \cap L^2$.

Homogeneous Besov spaces

By means of the Littlewood-Paley dyadic decomposition of unity

$$\{\varphi_j\}_{j \in \mathbb{Z}} : \hat{\varphi}(\xi) = \psi(2^{-j}\xi), \sum_{j \in \mathbb{Z}} \varphi_j(\xi) = 1 \ (\xi \neq 0)$$

with

$$\hat{\varphi}_j(\xi) = \psi(2^{-j}\xi) \quad \psi \in C_0^\infty(B_2 \setminus B_{\frac{1}{2}}),$$

where B_R is the ball in $\mathbb{R}^3$ centered at the origin with radius $R > 0$., for $s \in \mathbb{R}$, $1 \leq p, q \leq \infty$, we can define

$$\|f\|_{\dot{B}_{p,q}^s} := \begin{cases} \left(\sum_{j \in \mathbb{Z}} 2^{js} \|\varphi_j * f\|_{L^p}^q \right)^{\frac{1}{q}}, & \text{if } 1 < q < \infty, \\ \sup_{j \in \mathbb{Z}} \{ 2^{js} \|\varphi_j * f\|_{L^p} \}, & \text{if } q = \infty. \end{cases}$$

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In particular,

$$\dot{B}_{\infty,q}^0 \text{-norm} = \|f\|_{\dot{B}_{\infty,q}^0} := \begin{cases} \left(\sum_{j \in \mathbb{Z}} \|\varphi_j * f\|_\infty^q \right)^{\frac{1}{q}} & \text{if } 1 < q < \infty, \\ \sup_{j \in \mathbb{Z}} \{\|\varphi_j * f\|_{L^\infty}\}, & \text{if } q = \infty. \end{cases}$$

- We shall make use of **estimates by means of logarithmic growth of the H^s -norm**, $s > 3/2$ [Kozono-Ogawa-Taniuchi (2002)]:

$$\|f\|_{\dot{B}_{\infty,2}^0} \leq C \left(1 + \|f\|_{\dot{B}_{\infty,\infty}^0} \ln^{\frac{1}{2}}(e + \|f\|_{H^s}) \right).$$

The homogeneous Besov spaces $\dot{B}_{p,q}^s(\mathbb{R}^3)$ are defined to be

$$\dot{B}_{p,q}^s(\mathbb{R}^3) := \left\{ f \in \mathcal{S}'(\mathbb{R}^3) / \mathcal{P}(\mathbb{R}^3) : \|f\|_{\dot{B}_{p,q}^s} < \infty \right\}$$

where $\mathcal{S}'$ is the space of tempered distributions and $\mathcal{P}$ is the space of polynomials.

- $\dot{B}_{\infty,\infty}^0$ is quite larger than BMO : $f \in BMO$ (Bounded Mean Oscillation space) if $f \in L_{\text{loc}}^1$ and

$$[f]_Q := \int_Q \left| f(x) - \int_Q f \right| dx \text{ is uniformly bounded, on any cube } Q.$$

- BMO is larger than $\dot{B}_{\infty,2}^0$.

BMO-estimates of L^p -products [Kozono-Taniuchi (2000)]. Let $p \in (1, \infty)$; for all $f, g \in L^p \cap BMO$, there exists $C = C(p) > 0$:

$$\|f \cdot g\|_p \leq C(\|f\|_p \|g\|_{BMO} + \|f\|_{BMO} \|g\|_p).$$

The main result

Assume that

- ▶ $3 < \alpha, \beta < 4$; $u_0, v_0, \theta_0 \in H^s$, for $3/2 < s \leq 2$, with $\operatorname{div} u_0 = 0$.
- ▶ Let (u, v, θ) be a local solution of the system (1), defined on some time interval $[0, T)$, with $0 < T < \infty$, and such that, for any $0 < \tilde{T} < T$,

$$u, v, \theta \in C([0, \tilde{T}]; H^s) \quad \text{and} \quad u, v \in L^2(0, \tilde{T}; H^{s+1}), \quad (2)$$

where we used the notation of **Bochner spaces**.

Theorem 1 ([B.-Bisconti-Catania 2022])

The local solution $(u, v, \theta)(t)$ can be extended beyond time T , with the same regularity as in (2), and hence as a smooth solution, provided that

$$\int_0^T (\|u(t)\|_{\dot{B}_{\infty,\infty}^\delta}^\delta + \|v(t)\|_{\dot{B}_{\infty,\infty}^\gamma}^\gamma) dt < \infty, \quad (3)$$

where

$$\delta = \frac{6(\alpha - 1)}{3\alpha - 5} > 2 \quad \text{and} \quad \gamma = \frac{6(\beta - 1)}{3\beta - 5} > 2.$$

Comments on Theorem 1

- ▶ Theorem 1 highlights (somehow) that **without $\alpha \geq 4$** suitable constraints seems to be necessary in order to obtain regular global solutions.
- ▶ The fact that under (2), **solutions are smooth** follow by **[Fan, Wu (2019)]**.
- ▶ Regularity criteria in terms of homogeneous Besov spaces already in a number of recent publications (see e.g. **[Li, Zhai, Yin (2019)]** for 2D-TCM, **[Gala-Ragusa (2016), Dao-Díaz (2021)]** for 3D NS, **[Guo-Zhang (2020)]** for 3D MHD).

Sketches of the proof of Theorem 1

- ▶ The proof consists in proving **suitable energy estimates**, showing explicitly:

$$\|\Lambda^s(u, v, \theta)(T)\|^2 = \|\Lambda^s u\|^2 + \|\Lambda^s v\|^2 + \|\Lambda^s \theta\|^2 < +\infty, \quad (4)$$

where Λ^s is the fractional Laplacian with symbol $|\xi|^{2s}$, i.e.

$$\widehat{(-\Delta)^s f} = |\xi|^{2s} \hat{f}.$$

- ▶ In order to obtain (4), we need to establish **low-order estimates** in L^2 , L^{2m} , for $m > 1$, and H^1 , preliminarily.
- ▶ **Main tools (inequalities in Sobolev and interpolation spaces):**
 - ▶ standard Gagliardo-Nirenberg's, Hölder's, Young's,
 - ▶ specific estimates in terms of **BMO** and **logarithmic Sobolev inequality in Besov spaces** (as far as concerned low-order estimates)
 - ▶ **s-product estimates in H^s** (as far as concerned the higher-order estimates).

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L^2 -estimates

Taking the L^2 -inner product of $(1)_1$, $(1)_2$ and $(1)_3$ with u , v and θ , respectively, adding them up and integrating with respect to t , we reach

$$\|(u, v, \theta)(t)\|^2 + 2 \int_0^t \left(\|\nabla(u, v, \theta)\|^2 + \|u\|_{\alpha+1}^{\alpha+1} + \|v\|_{\beta+1}^{\beta+1} \right) = \|(u_0, v_0, \theta_0)\|^2,$$

where $\|\cdot\| = \|\cdot\|_2$.

We note that the role of the damping terms is to lead to

$$u \in L^{\alpha+1}(0, T; L^{\alpha+1}) \quad \text{and} \quad v \in L^{\beta+1}(0, T; L^{\beta+1}).$$

L^{2m} -estimates, $m > 1$

Differently from above, the term $\nabla\pi$ now gives a (not necessarily null) contribute:

$$\int_{\mathbb{R}^3} \nabla\pi \cdot |u|^{2m-2} u \, dx.$$

To handle with it, we essentially apply the *Leray projection* (projection on divergence-free vector fields)

$$w \mapsto w + \nabla(-\Delta)^{-1}(\operatorname{div} w)$$

to $(1)_1$, where $(-\Delta)^{-1}$ is the pseudo-differential operator whose associated Fourier symbol is given by $-1/|\xi|^2$, to obtain

$$\begin{aligned} u_t + (u \cdot \nabla)u - \Delta u + |u|^{\alpha-1}u = \\ - \operatorname{div}(v \otimes v) - \nabla(-\Delta)^{-1}\operatorname{div}(|u|^{\alpha-1}u) \\ - \nabla(-\Delta)^{-1}\operatorname{div}(\operatorname{div}(u \otimes u + v \otimes v)). \end{aligned} \tag{5}$$

By multiplying both sides of (5) by $|u|^{2m-2}u$, we obtain

$$\begin{aligned}
 & \frac{d}{dt} \|u\|_{2m}^{2m} + \|u\|_{2m+\alpha-1}^{2m+\alpha-1} + \text{positive terms} = \\
 & - \int_{\mathbb{R}^3} \operatorname{div}(v \otimes v) \cdot |u|^{2m-2} u \, dx \\
 & - \int_{\mathbb{R}^3} \nabla(-\Delta)^{-1} \operatorname{div}(|u|^{\alpha-1} u) \cdot u |u|^{2m-2} \, dx \\
 & - \int_{\mathbb{R}^3} \nabla(-\Delta)^{-1} \operatorname{div}(\operatorname{div}(u \otimes u + v \otimes v)) \cdot u |u|^{2m-2} \, dx \\
 & \leq C(\|u\|_{2m}^{2m} + \|v\|_{2m}^{2m}) (1 + (\|u\|_{\dot{B}_{\infty,\infty}^0}^2 + \|v\|_{\dot{B}_{\infty,\infty}^0}^2) \ln(e + \|u\|_{H^s} + \|v\|_{H^s})) \\
 & + \text{terms to be reabsorbed in the positive terms on the left-hand side} \\
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By using BMO estimates of $2m$ -products and the logarithmic inequality in terms of Besov spaces.

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\end{aligned}$$

By multiplying both sides of (5) by $|u|^{2m-2}u$, we obtain

$$\begin{aligned} & \frac{d}{dt} \|u\|_{2m}^{2m} + \text{positive terms} \\ & \leq C(\|u\|_{2m}^{2m} + \|v\|_{2m}^{2m}) (1 + (\|u\|_{\dot{B}_{\infty,\infty}^0}^2 + \|v\|_{\dot{B}_{\infty,\infty}^0}^2) \ln(e + \|u\|_{H^s} + \|v\|_{H^s})). \end{aligned}$$

Summing to the above inequality those coming from multiplying $(1)_2$ and $(1)_3$ by (resp.) $|v|^{2m-2}v$ and $|\theta|^{2m-2}\theta$, yields

$$\begin{aligned} & \frac{d}{dt} \|(u, v, \theta)(t)\|_{2m}^{2m} + \text{positive terms} \\ & \leq C \|(u, v, \theta)\|_{2m}^{2m} (1 + (\|u\|_{\dot{B}_{\infty,\infty}^0}^2 + \|v\|_{\dot{B}_{\infty,\infty}^0}^2) \ln(e + \|u\|_{H^s} + \|v\|_{H^s})), \end{aligned}$$

with C do not depend on (u, v, θ) .

Remark 2

In the **positive terms**, the addendum $\|v\|_{2m+\beta-1}^{2m+\beta-1}$ appears explicitly, while $\|u\|_{2m+\alpha-1}^{2m+\alpha-1}$ is used to handle with $\nabla\pi$, as already observed.

By applying **Gronwall's inequality**, for any $t < T$ and $\varepsilon > 0$, we deduce

$$\begin{aligned}
 & \| (u, v, \theta)(t) \|_{2m}^{2m} + \int_{t_*}^t \text{positive terms} \\
 & \leq \| (u, v, \theta)(t_*) \|_{2m}^{2m} e^{C \int_{t_*}^t \left(\|u\|_{\dot{B}_{\infty, \infty}^0}^2 + \|v\|_{\dot{B}_{\infty, \infty}^0}^2 \right) \ln(e + \|u\|_{H^s} + \|v\|_{H^s}) d\ell} \\
 & \leq C_* \left(e + \sup_{[t_*, t]} \| (u, v, \theta) \|_{H^s} \right)^{C\varepsilon},
 \end{aligned}$$

where C_* depends only on $\| (u, v, \theta)(t_*) \|_{2m}^{2m}$, $0 < t_* < t$ is such that

$$\int_{t_*}^t \left(\|u\|_{\dot{B}_{\infty, \infty}^0}^2 + \|v\|_{\dot{B}_{\infty, \infty}^0}^2 \right) < \varepsilon.$$

The integral condition (3) implies that for every $\varepsilon > 0$ and $0 \leq t < T$ there exists $0 < t_* < t$ such that

$$\int_{t_*}^t \left(\|u\|_{\dot{B}_{\infty, \infty}^0}^{\delta} + \|v\|_{\dot{B}_{\infty, \infty}^0}^{\gamma} \right) < \varepsilon, \text{ and hence } \int_{t_*}^t \left(\|u\|_{\dot{B}_{\infty, \infty}^0}^2 + \|v\|_{\dot{B}_{\infty, \infty}^0}^2 \right) < \varepsilon.$$

L^{2m} -estimates in terms of H^s

For every $0 \leq t_* \leq t < T$, we obtained

$$\|(u, v, \theta)(t)\|_{2m}^{2m} \leq C_* \left(e + \sup_{[t_*, t]} \|(u, v, \theta)\|_{H^s} \right)^{C\varepsilon}.$$

By explicit write down the integral terms in of the positive terms in the left-hand side, we notice that they are involved:

$$\begin{aligned} v &\in L^{2m+\beta-1}(t_*, t; L^{2m+\beta-1}) \\ |u|^{m-1} \nabla u, |v|^{m-1} \nabla v, |\theta|^{m-1} \nabla \theta &\in L^2(t_*, t; L^2) \end{aligned}$$

Remark 3

Our procedure does not provide the regularity information of $u \in L^{2m+\alpha-1}(t_*, t; L^{2m+\alpha-1})$ because the damping term in u has been used to handle the gradient of pressure $\nabla \pi$.

$\dot{H}^1$ -estimates

By multiplying $(1)_1$, $(1)_2$ and $(1)_3$ by (resp.) $-\Delta u$, $-\Delta v$ and $-\Delta \theta$, integrating by parts, and summing up them, we get

$$\begin{aligned} & \frac{d}{dt} \|\nabla(u, v, \theta)(t)\|^2 + \text{positive terms} \\ & \leq C \|\nabla(u, v, \theta)\|^2 \left(1 + \mathcal{A} + (\mathcal{A} + \|v\|_{\dot{B}_{\infty, \infty}^0}^\gamma + \|u\|_{\dot{B}_{\infty, \infty}^0}^\delta) \times \ln(e + \|u\|_{H^s} + \|v\|_{H^s}) \right) \end{aligned}$$

where

$$\mathcal{A} := \|u\|_{\alpha+1}^{\alpha+1} + \|v\|_{\beta+1}^{\beta+1}, \quad \gamma = \frac{6(\beta-1)}{3\beta-5}, \quad \delta = \frac{6(\alpha-1)}{3\alpha-5}.$$

The above differential inequality is very similar to that coming from L^{2m} -estimates. The main difference is the appearance of exponents $\gamma, \delta > 2$.

Where do $\gamma, \delta > 2$ come from?

Let us justify the introduction of γ :

Note, on the right-hand side, by Holder, Young, Gagliardo Nirenberg and interpolation inequalities, we obtain

$$\begin{aligned} \int_{\mathbb{R}^3} \operatorname{div}(v \otimes v) \cdot \Delta u \, dx &\leq C \int_{\mathbb{R}^3} |v| |\nabla v| |\Delta u| \\ &\leq C \|v\|_3^2 \|v\|_{BMO}^2 \|\nabla v\|^2 + \varepsilon \|\Delta v\|^2 + \varepsilon \|\Delta u\|^2 \\ &\leq C \|v\|_{\beta+1}^{\frac{2(\beta+1)}{3(\beta-1)}} (1 + \|v\|_{\dot{B}_{\infty,\infty}^0}^2 \ln(e + \|v\|_{H^s})) \|\nabla v\|^2 + \varepsilon \|\Delta v\|^2 + \varepsilon \|\Delta u\|^2, \end{aligned}$$

and, by Young's inequality,

$$\|v\|_{\beta+1}^{\frac{2(\beta+1)}{3(\beta-1)}} \|v\|_{\dot{B}_{\infty,\infty}^0}^2 \leq C \|v\|_{\beta+1}^{\beta+1} + \|v\|_{\dot{B}_{\infty,\infty}^0}^{2\frac{3(\beta-1)}{3\beta-5}}$$

$\dot{H}^1$ -estimates in terms of H^s

We can then repeat the same argument used in the case of L^{2m} -estimates, to get, for every $0 \leq t_* \leq t < T$,

$$\begin{aligned} & \|\nabla(u, v, \theta)(t)\|^2 + \int_{t_*}^t \text{positive terms} \\ & \leq \|\nabla(u, v, \theta)(t_*)\|^2 e^{C \int_{t_*}^t (\|u\|_{\dot{B}_{\infty, \infty}^0}^\delta + \|v\|_{\dot{B}_{\infty, \infty}^0}^\gamma) \ln(e + \|u\|_{H^s} + \|v\|_{H^s}) d\ell} \\ & \leq C_* (e + \sup_{[t_*, t]} \|(u, v, \theta)\|_{H^s})^{C\varepsilon}, \end{aligned}$$

where y defined above and the multiplying constant still denoted by C_* is now $\|\nabla(u, v, \theta)(t_*)\|^2$.

Remark 4

The *positive terms* involve information on

$$\begin{aligned} \Delta u, \Delta v, \Delta \theta & \in L^2(t_*, t; L^2), \\ |u|^{\frac{\alpha-1}{2}} \nabla u, |v|^{\frac{\beta-1}{2}} \nabla v & \in L^2(t_*, t; L^2). \end{aligned}$$

$\dot{H}^s$ -estimates, useful instrumental estimates

[s-product-estimates] for $s > 0$, let

$$\Lambda^s = (-\Delta)^{\frac{s}{2}},$$

and $q_1, q_2 \in (0, \infty)$ such that $\frac{1}{p} = \frac{1}{p_1} + \frac{1}{q_1} = \frac{1}{p_2} + \frac{1}{q_2}$. We have:



$$\begin{aligned} \|\Lambda^s, f\|_p &= \|\Lambda^s(fg) - f\Lambda^s g\|_p \\ &\leq C(\|\nabla f\|_{p_1} \|\Lambda^{s-1} g\|_{q_1} + \|g\|_{p_2} \|\Lambda^s f\|_{q_2}), \end{aligned}$$



$$\|\Lambda^s(fg)\|_p \leq C(\|f\|_{p_1} \|\Lambda^s g\|_{q_1} + \|g\|_{p_2} \|\Lambda^s f\|_{q_2}).$$

$\dot{H}^s$ -estimates, $3/2 < s \leq 2$

We apply the operator Λ^s to each of $(1)_1$, $(1)_2$ and $(1)_3$ and multiply the resulting equation in L^2 , respectively, by $\Lambda^s u$, $\Lambda^s v$ and $\Lambda^s \theta$. In performing the next calculations we take $m > 3(\max\{\alpha, \beta\} - 2)$.

$$\begin{aligned} & \frac{1}{2} \frac{d}{dt} \|\Lambda^s(u, v, \theta)\|^2 + \frac{1}{2} [\|\Lambda^{s+1} u\|^2 + \|\Lambda^{s+1} v\|^2 + \|\Lambda^{s+1} \theta\|^2] \\ & \leq C (\|(u, v)\|_{2m}^{2m})^{\Gamma_1} \|\nabla(u, v)\|^{\Gamma_2} (1 + \|\Delta(u, v)\|^2) \\ & \quad + C (\|(u, v)\|_{2m}^{2m})^{\frac{(2s-1)2s}{(2m-3)(2s-3)}} + C \|\nabla(u, v)\|^{\frac{2\bar{\sigma}}{\bar{\sigma}-1}} + C \|\Delta(u, v)\|^2 \\ & \quad + C \|\Lambda^s(u, v, \theta)\|^2 \\ & =: \mathcal{A}_1 + \mathcal{A}_2 + C \|\Lambda^s(u, v, \theta)\|^2, \end{aligned}$$

for some $C, \Gamma_1, \Gamma_2 > 0$ and $\bar{\sigma} > 1$ given implicitly.

By Gronwall's inequality, we obtain

$$\begin{aligned} & (e + \|\Lambda^s(u, v, \theta)(t)\|^2) \\ & \leq \left((e + \|\Lambda^s(u, v, \theta)(t_*)\|^2) + \int_{t_*}^t \mathcal{A}_1 \, d\ell + \int_{t_*}^t \mathcal{A}_2 \, d\ell \right) e^{C(t-t_*)}. \end{aligned}$$

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We have

$$\begin{aligned}
\int_{t_*}^t \mathcal{A}_1 d\ell &= C \int_{t_*}^t (\|(u, v)\|_{2m}^{2m})^{\Gamma_1} \|\nabla(u, v)\|^{\Gamma_2} (1 + \|\Delta(u, v)\|^2) d\ell \\
&\leq C \sup_{t_* \leq \ell \leq t} (\|(u, v)(\ell)\|_{2m}^{2m})^{\Gamma_1} \|\nabla(u, v)(\ell)\|^{\Gamma_2} \int_{t_*}^t (1 + \|\Delta(u, v)\|^2) d\ell \\
&\leq C_*(e + y(t))^{C\Gamma_1\epsilon} (e + y(t))^{C\Gamma_2\epsilon} ((t - t_*) + C_*(e + y(t))^{C\epsilon}) \\
&\leq C(1 + (t - t_*))(e + y(t))^{C(\Gamma_1 + \Gamma_2 + 1)\epsilon},
\end{aligned}$$

where

$$y(t) := \sup_{[t_*, t]} \|(u, v, \theta)\|_{H^s}.$$

Similarly,

$$\int_{t_*}^t \mathcal{A}_2 d\ell \leq C(1 + (t - t_*))(e + y(t))^{C(1+\Sigma)\epsilon}.$$

Therefore, we obtain

$$\begin{aligned} & (e + \|\Lambda^s(u, v, \theta)(t)\|^2) \\ & \leq C (e + \|\Lambda^s(u, v, \theta)(t_*)\|^2 + (t - t_*)) e^{C(t-t_*)} (e + y(t))^{C\Theta\varepsilon}, \end{aligned}$$

where $\Theta > 0$ is a positive real number.

Taking the supremum over $0 \leq \ell \leq t$, we have

$$\begin{aligned} (e + y(t)^2) & \leq (e + \|\Lambda^s(u, v, \theta)(t_*)\|^2 + (t - t_*)) (e + y(t))^{C\Theta\varepsilon} e^{C(t-t_*)} \\ & \leq (e + \|\Lambda^s(u, v, \theta)(t_*)\|^2 + (t - t_*)) e^{C(t-t_*)} (e + y(t)^2)^{C\Theta\varepsilon}. \end{aligned}$$

Also, taking t_* sufficiently close to t so that ε is such that $C\Theta\varepsilon < 1$ we conclude that

$$(e + y(t)^2)^{1-C\Theta\varepsilon} \leq (e + \|\Lambda^s(u, v, \theta)(t_*)\|^2 + (t - t_*)) e^{C(t-t_*)} < +\infty.$$

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We finally close the $\dot{H}^s$ -estimates

Hence,

$$\sup_{t_* \leq \ell \leq t} (e + \|\Lambda^s(u, v, \theta)(\ell)\|^2) < +\infty \quad \text{for every } t \in (0, T).$$

Since t is arbitrary in the interval $(0, T)$, we have also that

$$\|\Lambda^s(u, v, \theta)(T)\|^2 < +\infty,$$

and this concludes the proof of Theorem 1.

Further results

Next steps could be study a **temperature-dependent viscous term** in (TCM):

- ▶ Introduce, other than damping, a temperature-dependent viscous term, i.e. a term of the type

$$-\operatorname{div}(\mu(\theta)\nabla u) \text{ in place of } -\Delta u,$$

the same for $(1)_1$ and $(1)_2$ and some other regularization term, as the Voigt-regularization, that is

$$\text{replace } \partial_t u \text{ with } \partial_t(I - \alpha^2 \Delta)u.$$

- ▶ In the literature has been **only studied the 2D case** (see **Ye, Zhu (2020)**, **Dong, Li, Xu, Ye (2021)**, and **Li, Xu, Ye (2022)**).

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Thanks for your kind attention!

Strong solutions

A solution u is a global *strong solution* if, for every $T > 0$

$$L^\infty(0, T; H^1) \cap L^2(0, T; H^2).$$

[Fan (2019)]-TCM without damping

If $\nabla u \in L^q(0, T; L^p(\mathbb{R}^3))$ with

$$\frac{3}{p} + \frac{2}{q} \leq 2, \quad 2 < p \leq 3 \tag{6}$$

then u is smooth in $[0, T] \times \mathbb{R}^3$.

Since, with $3/2 < s \leq 2$, $W^{s,2}$ is embedded in $W^{1,p}$, with $p \geq 3/2$, and by choosing $q = \infty$ then $u \in C([0, \tilde{T}]; H^s)$ implies (6).